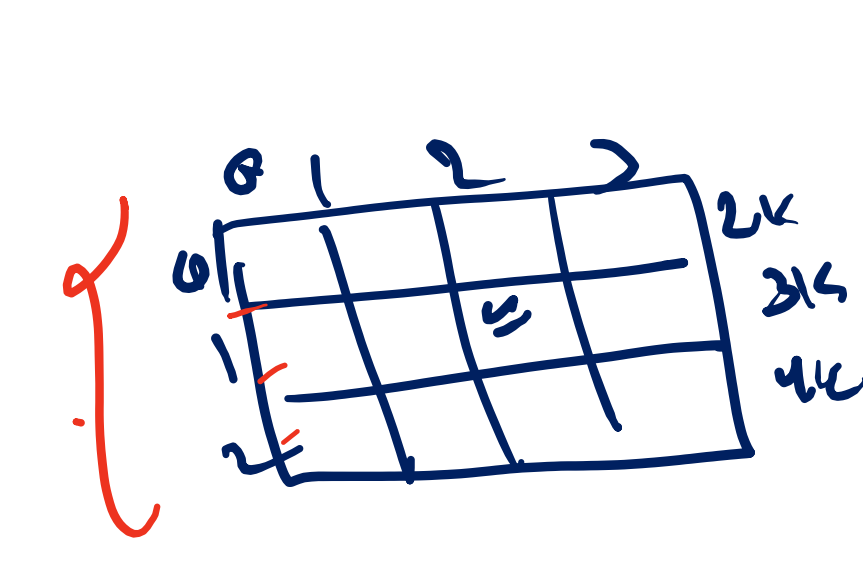
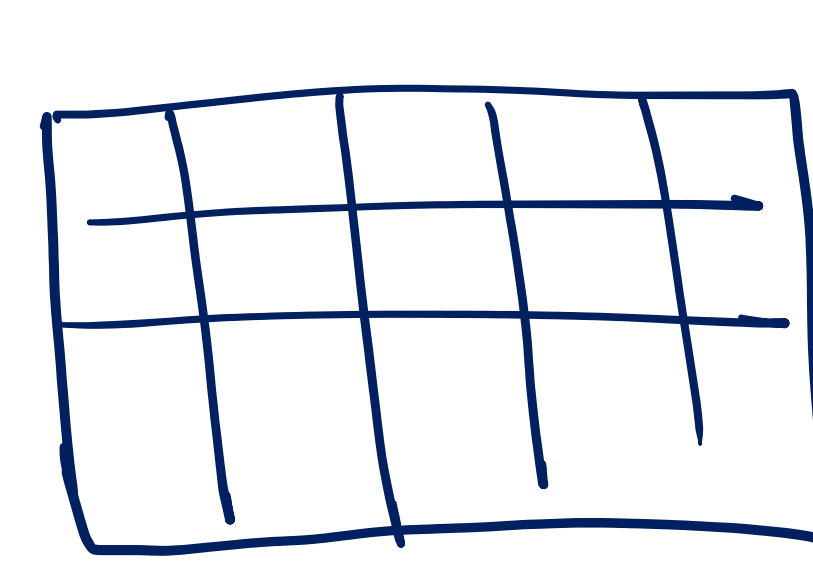


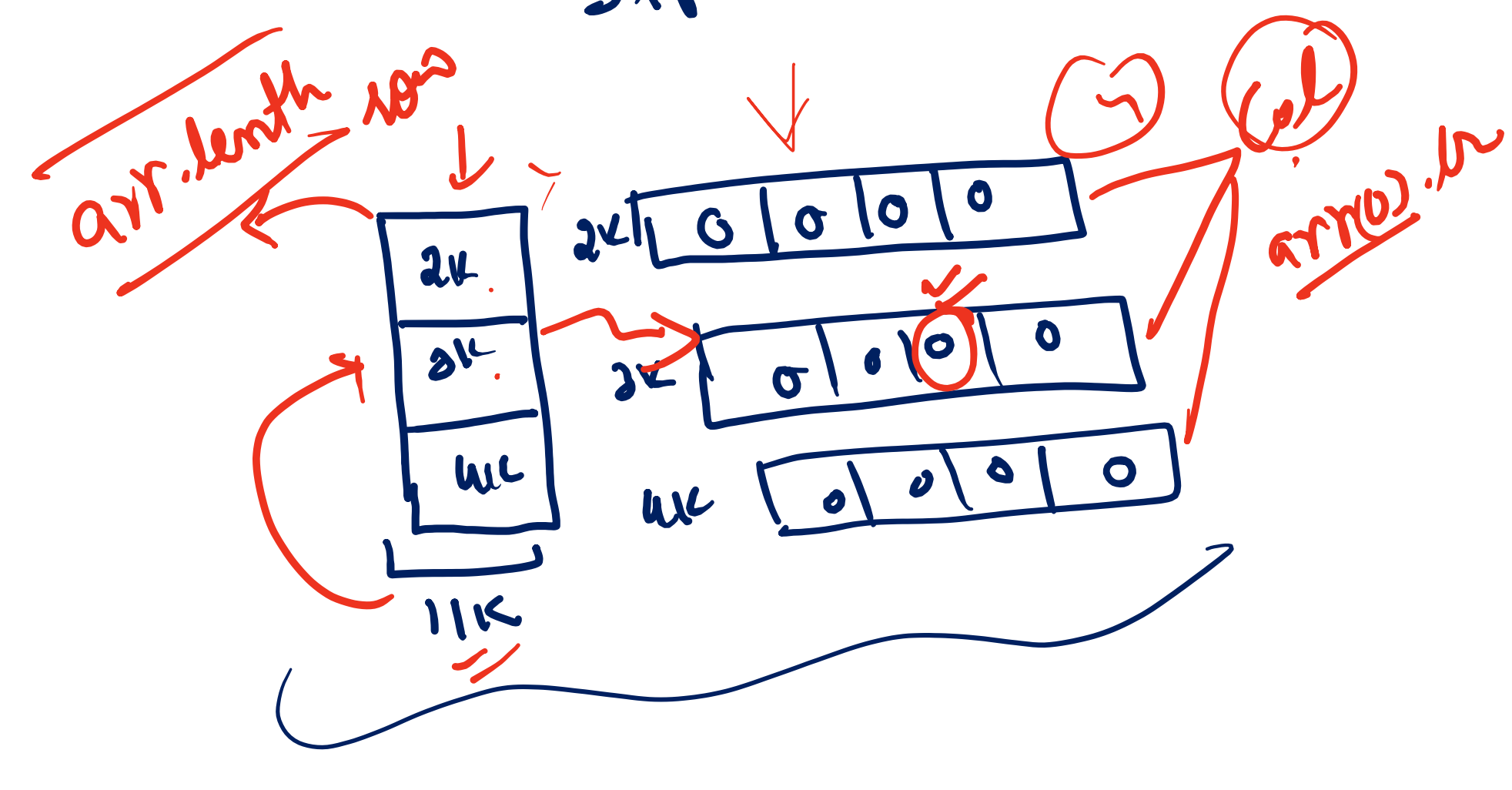
2D Array



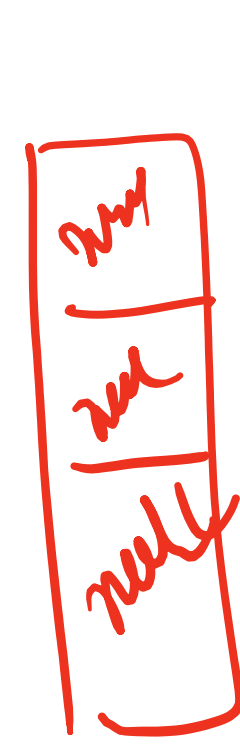
3x4

```
int [] [] arr = new int [3][4];
```

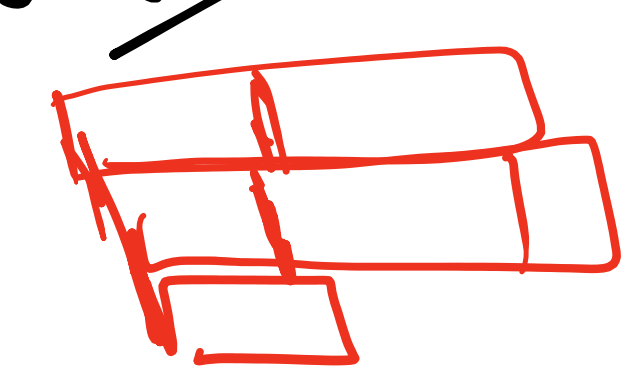
arr(1)(2)
34



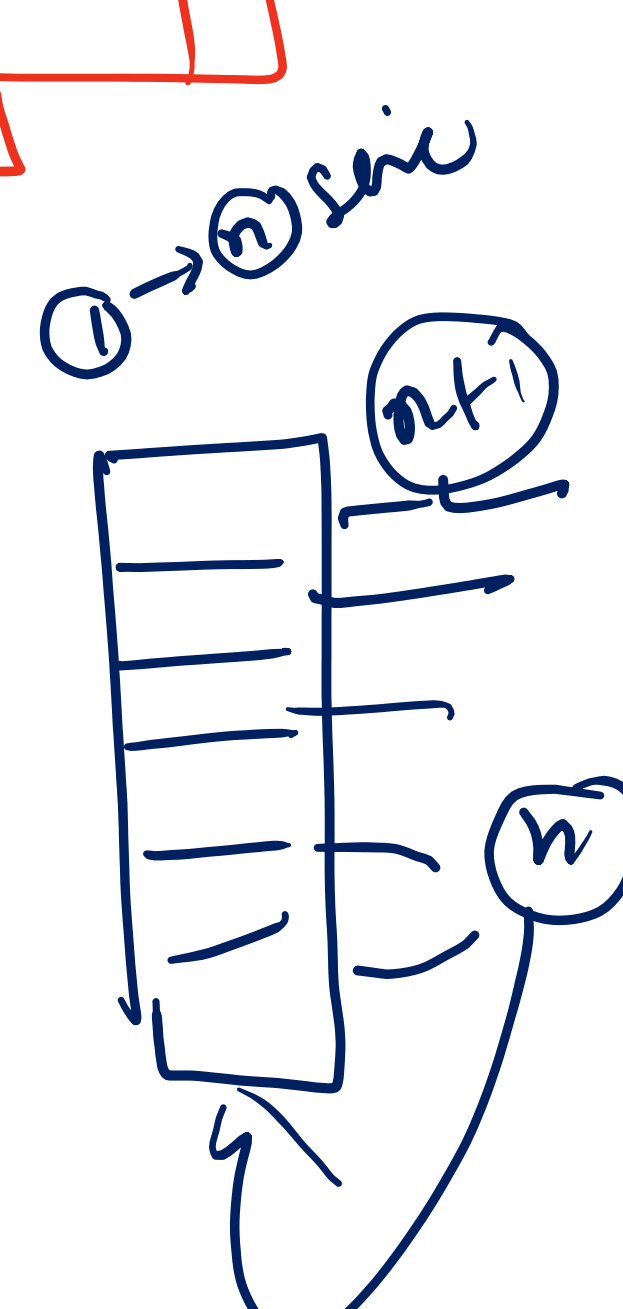
```
int [] [] arr = new int [3][];  
arr[0] = new int [2];  
arr[1] = new int [4];  
arr[2] = new int [3];
```



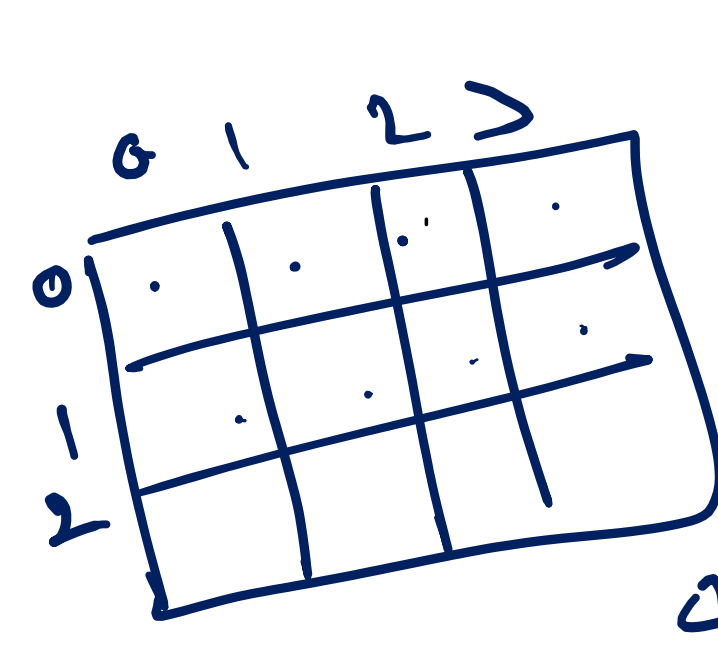
Jagged



```
int[] [] arr = new int [n][m];  
a) n+1  
b) m  
c) m+1  
d) n
```



Positioning



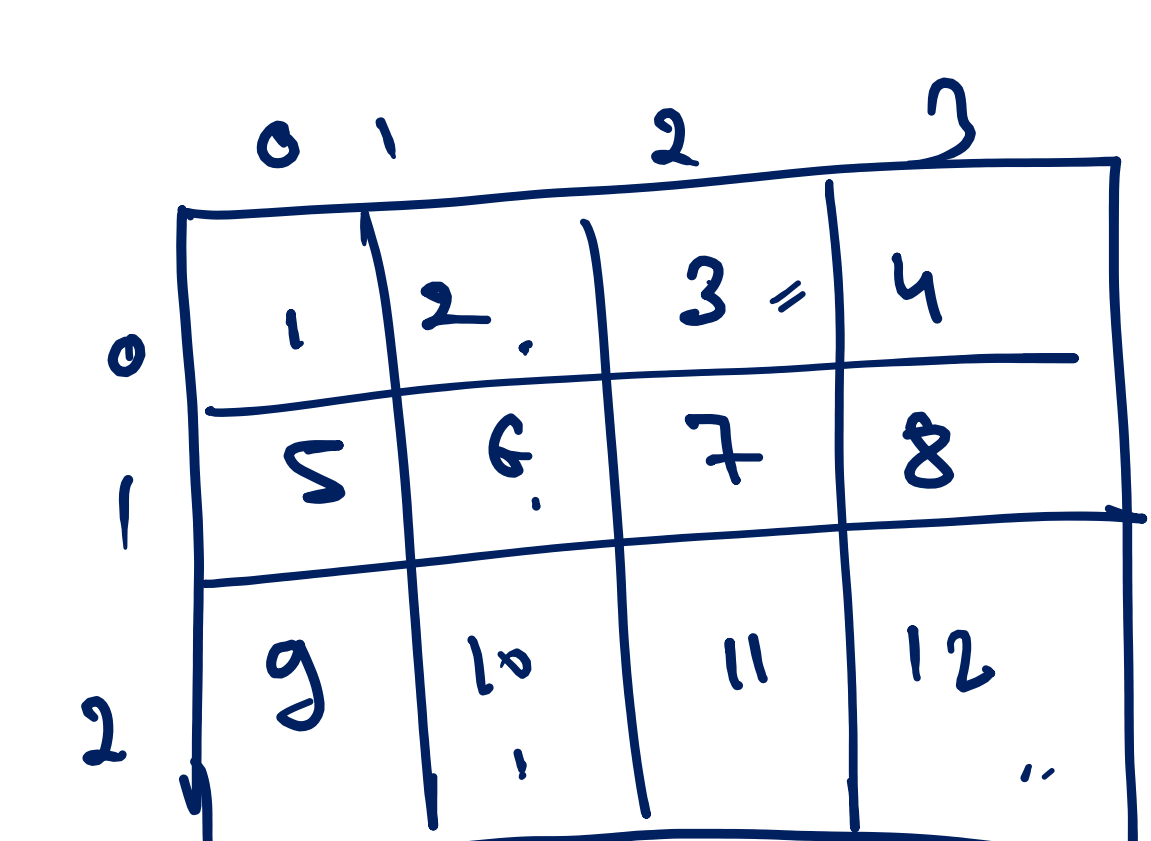
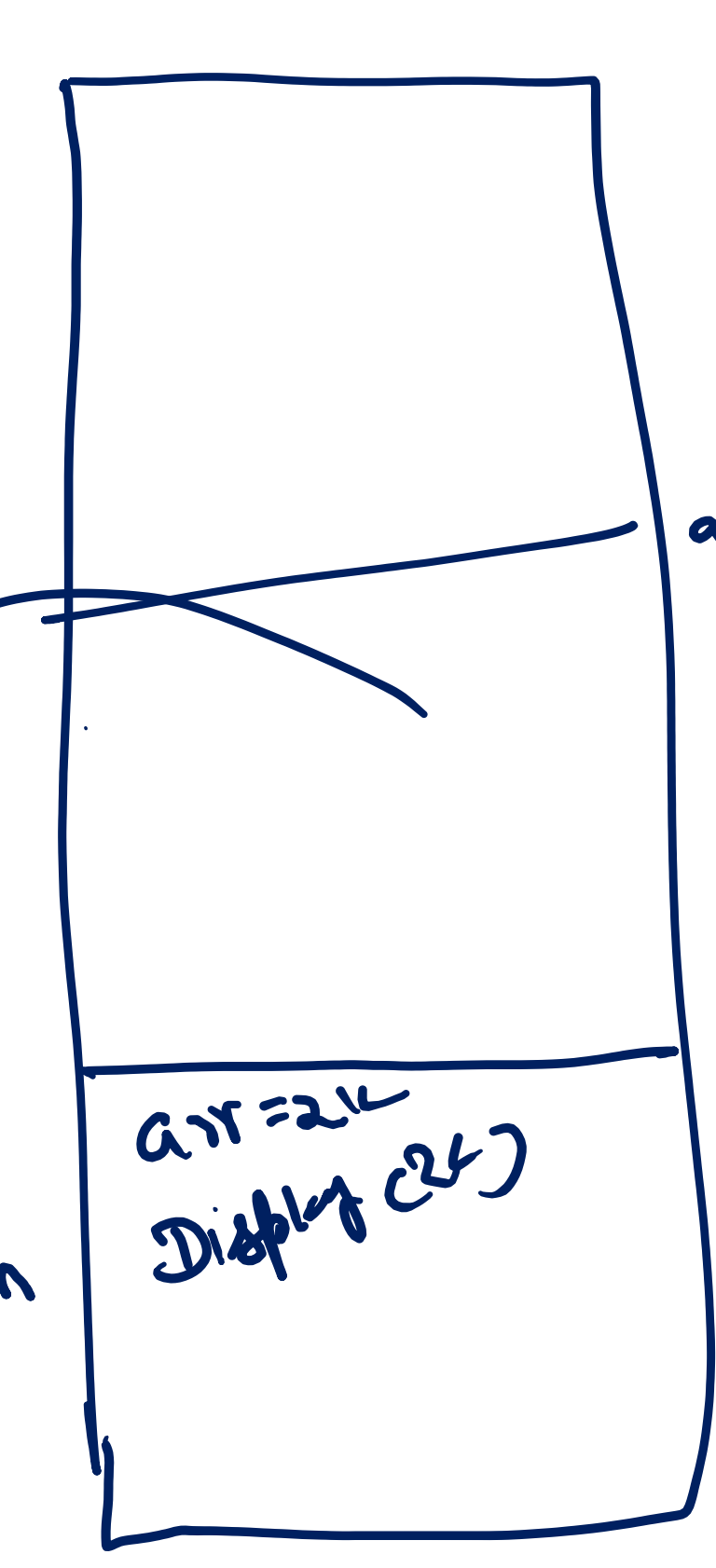
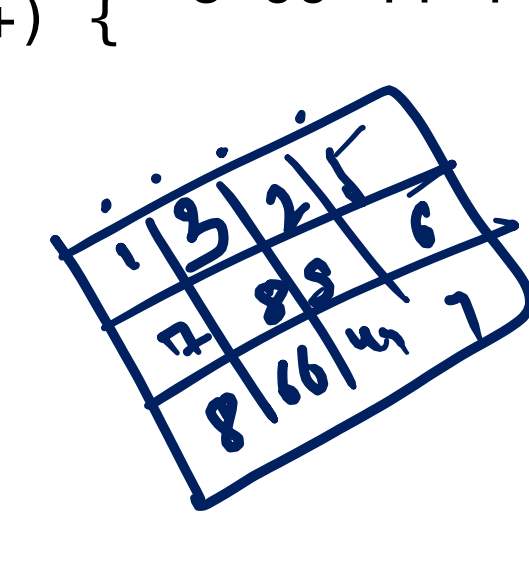
Posci = i * arr[0].length + j
Posci = 0 * 4 + 0 = 0
Posci = 1 * 4 + 0 = 4
Posci = 2 * 4 + 0 = 8

Row i, Col j
0-0 1-0 2-0
0-1 1-1 2-1
0-2 1-2 2-2
0-3 1-3 2-3

arr[i][j] = scanner.nextInt();

```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    Scanner sc = new Scanner(System.in);  
    int n = sc.nextInt();  
    int m = sc.nextInt();  
    int [] [] arr = new int [n][m];  
    for (int i = 0; i < arr.length; i++) {  
        for (int j = 0; j < arr[0].length; j++) {  
            arr[i][j] = sc.nextInt();  
        }  
    }  
    Display(arr);  
    public static void Display(int [] [] arr) {  
        for (int i = 0; i < arr.length; i++) {  
            for (int j = 0; j < arr[0].length; j++) {  
                System.out.print(arr[i][j] + " ");  
            }  
            System.out.println();  
        }  
    }  
}
```

3 4
1 3 2 5
7 8 9 6
8 66 44 4
1 3 2 5
7 8 9 6
8 66 44 4



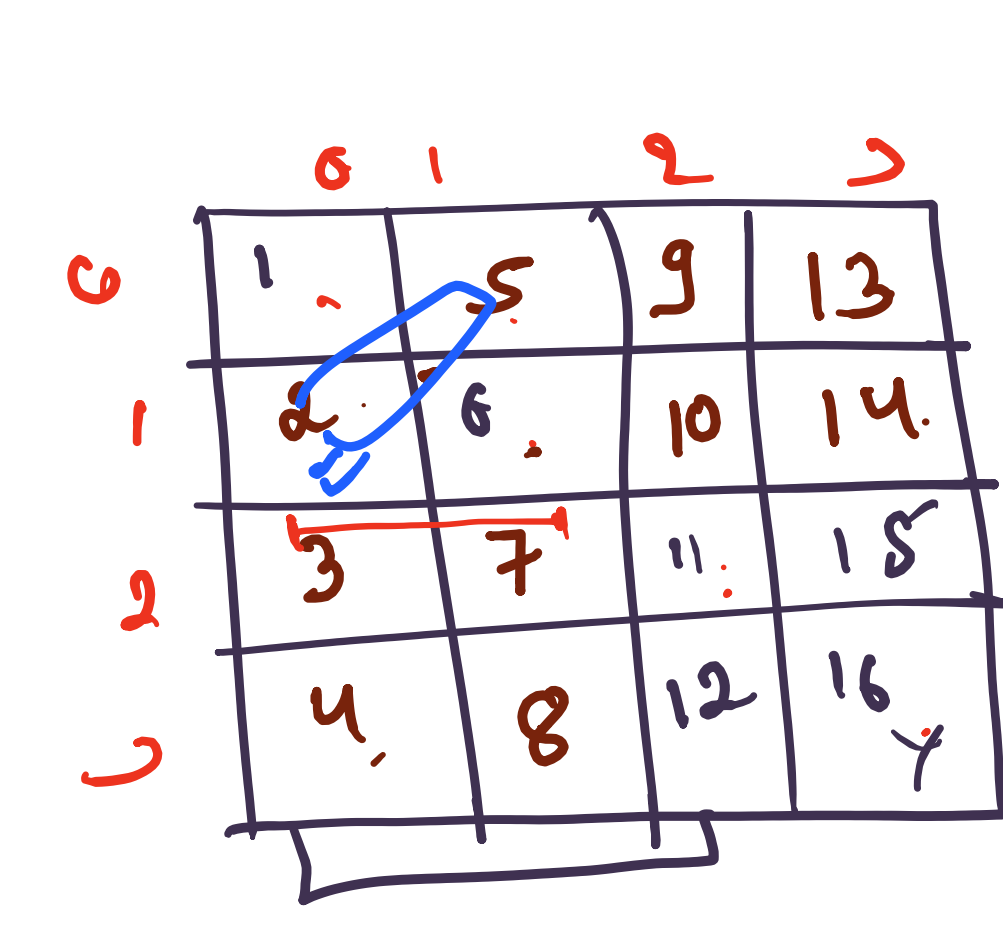
0-0 1-0 2-0
0-1 1-1 2-1
0-2 1-2 2-2
0-3 1-3 2-3

arr[i][j]

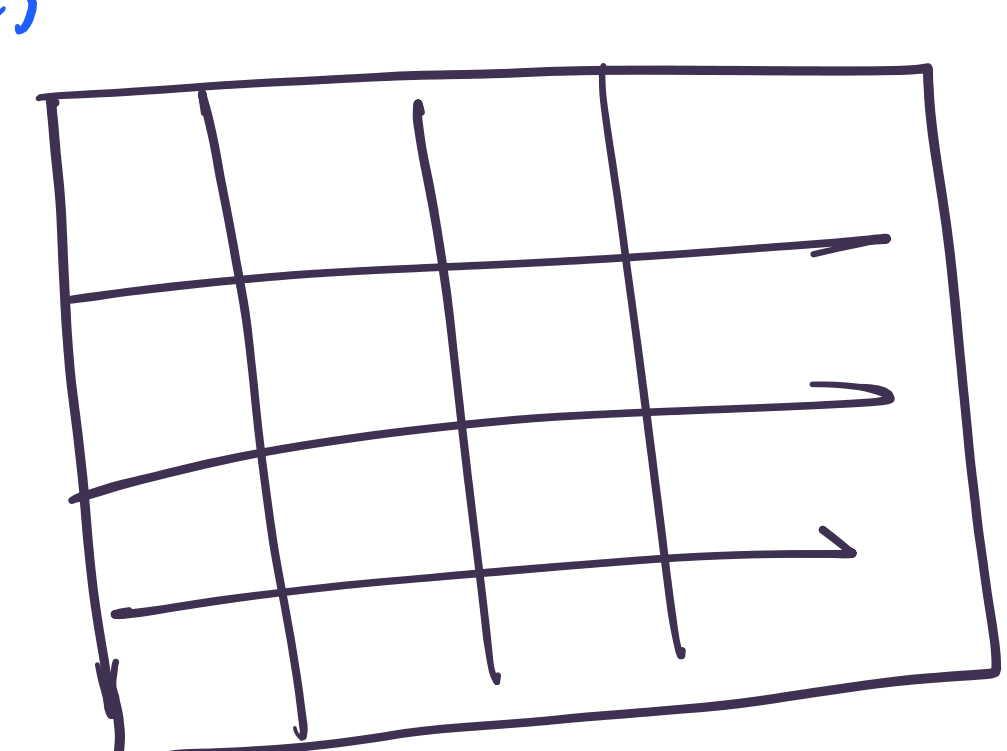
1 5 9 10 6 2 3 7 11 12 8 4

for (col = 0; col < arr[0].length; col++)
 if (col % 2 == 0)
 for (row = 0; row < arr.length; row++)
 System.out.print(arr[row][col]);
 else
 for (row = arr.length - 1; row >= 0; row--)
 System.out.print(arr[row][col]);

(i,j) -> (i,j)



0-0 1-0



Posci = i * arr[0].length + j
Posci = 0 * 4 + 0 = 0
Posci = 1 * 4 + 0 = 4
Posci = 2 * 4 + 0 = 8
Posci = 3 * 4 + 0 = 12

0-0 1-0
0-1 1-1
0-2 1-2
0-3 1-3
1-0 2-0
1-1 2-1
1-2 2-2
1-3 2-3

2-0 3-0
2-1 3-1
2-2 3-2
2-3 3-3

1	4	7	11	15
2	5	8	12	19
3	6	9	16	22
10	13	14	17	24
18	21	23	26	30

15 > 13

item = 13

n * log(m)

row = 0, col = arr[0].length - 1
while (row <= 0 & col >= 0)
 if (arr[row][col] == item)
 return true
 else if (arr[row][col] > item)
 col--
 else
 row++